Supplementary Information

This supplement gives the implementation details and sensitivity results that are needed to reproduce the study but are not needed to follow the main paper. It covers the semantic model roles, panorama admission, numerical settings, provenance, convergence, the material-evidence control, timing boundaries, and additional limitations. All numerical statements refer to the verified five-site calculations under the first-material-interaction transport contract.

I. Semantic evidence and surface fusion

The semantic pass uses two models with separate roles. Mask2Former with the 65-class Mapillary Vistas vocabulary assigns one street-scene entity to every pixel. It therefore supplies the complete entity layer and identifies small objects such as poles, signs, curbs, and bicycle racks. One Vistas building label can contain brick, stone, render, glass, and metal. SAM 3 supplies the open-vocabulary material evidence needed inside such broad entity classes. The production catalog contains 60 text prompts and 61 raster identifiers including the unlabeled identifier. Examples include “brick facade”, “glass window”, “metal cladding panel”, and separate ground, grass, shrub, tree, and forest concepts. The prompts are also gated by the Vistas classes. A class must occupy at least 256 pixels in a 1536×1536 crop before its related prompts are sent to SAM 3. The gate reduces work and removes prompts that have no entity support in the view. The complete prompt list and its entity, material, attribute, and vegetation mappings are stored in `config/semantic_concepts.json`.

The four rectilinear crops of each panorama use yaw angles of $0°$, $90°$, $180°$, and $270°$, zero pitch, a $90°$ field of view, and 1536 pixels per side. Mask2Former runs at 1536 pixels. SAM 3 uses its trained 1008-pixel input, a score threshold of 0.35, and prompt batches of 32. Production fixes the full model, source, and catalog identities listed in Table I. It refuses another model pair or a catalog change, even if the number of prompts remains the same.

Panorama rays intersect the original support mesh. A common 8×8 barycentric atlas is defined on each observed support triangle. Its coordinate system is fixed to the triangle and is independent of the camera. Each camera first reduces all of its rays in one atlas cell to at most one confidence-weighted contribution. Camera means are then added across views. Range and raw pixel density give no extra weight. Therefore, two panoramas can place a material boundary at different positions on one large support triangle. Both observations are accumulated in the same barycentric cells and remain a distribution. The transport lookup uses the original triangle identifier and the barycentric coordinates of each ray hit. The highly tessellated mesh shown for atlas inspection is a display object and is not the transport mesh.

Material evidence remains probabilistic through fusion. Vistas-backed pixels use the declared full $p(\text{material} \mid \text{entity})$ table. Concept-backed pixels use the full material distribution of the detected prompt. Transport removes probability assigned to air, unknown material, people, vehicles, and participating volumes. It binds an image-derived structural interface only when the remaining compatible structural mass is strictly greater than 0.5. An exact tie and any unsupported atlas cell use the geometric face material. Reflected power uses the posterior-weighted material coefficients. The specular sampling probability uses the posterior-weighted reflected specular share. Grass keeps the geometric ground interface. Woody canopy evidence is nonblocking until registered closed canopy geometry can supply path chords for volume attenuation.

The per-view fishnet is a separate audit product. It cuts visible support triangles at semantic boundaries for inspection and stores each accepted piece with its source triangle, image support, confidence, area, and visible fraction. A parallel table keeps rejected candidate geometry when a reliable inverse projection exists. Reason codes include a degenerate or subpixel triangle, clipping outside the crop, occlusion by the support mesh, clutter in front, a transient object, missing semantic support, area below the retained minimum, a grazing plane, a pixel that is not owned by the support surface, and a mesh or pose conflict. A rejected piece can lie on the support mesh because the code describes why that image-space piece was not accepted as observed semantic evidence. Rejected pieces do not replace or remove the original transport geometry.

A. Ray-reached evidence coverage

The audit replays all five sealed routes and 16 seeds with the production geometry, source curve, atlas, body, 200,000 primary rays, and 4,096 output cells. It classifies the exact reflection point of every retained order-1 specular path and the first blocking material vertex of every accepted first-diffuse event. Category counts, transport contribution, body-coupled fields, and the complete replay all close to the sealed arrays. Direct transport is not assigned a category because it has no material interaction.

Panorama evidence informs most of the pooled non-direct body contribution because the exact specular term is both larger and more often atlas-bound. The first-diffuse component has a different support pattern. Most of its body-coupled contribution reaches geometry with no panorama evidence or with evidence refused by the host-material compatibility gate. The nonblocking woody category is zero for retained terminal interactions because a nonblocking crossing cannot be the first blocking material vertex. This zero does not imply that woody canopy evidence is absent from the routes.

II. Panorama registration and admission

The registration matches the segmented sky boundary in a panorama to the upper envelope of the support mesh. The stored residual is an angular skyline error. An admitted pose has a residual no greater than $4°$. The second test casts the directions labeled as sky from the fitted camera into the mesh. It records the fraction that hits geometry and the median range of those hits. A camera is classified as inside the geometry when the hit fraction is greater than 0.5 and the median range is less than 2 m. The paired condition matters because a camera inside a wall can still obtain a small silhouette residual. A large conflict at long range is retained as a distinct low-quality state.

TABLE I
IMMUTABLE SEMANTIC IDENTITIES USED BY THE PRODUCTION ATLAS.

Item	Immutable identifier
Mask2Former weights	`4772b6bf101d91f2534c106dc524d906aeb3c68a`
SAM 3 weights	`3c879f39826c281e95690f02c7821c4de09afae7`
SAM 3 source	`96914d2425f90a64f45ca977c2b5165418099543`
Parsed concept catalog	`66d0dfefba87bde5081cfa82108ca60d47c79cf641df3ec44129ec20bb90453b`

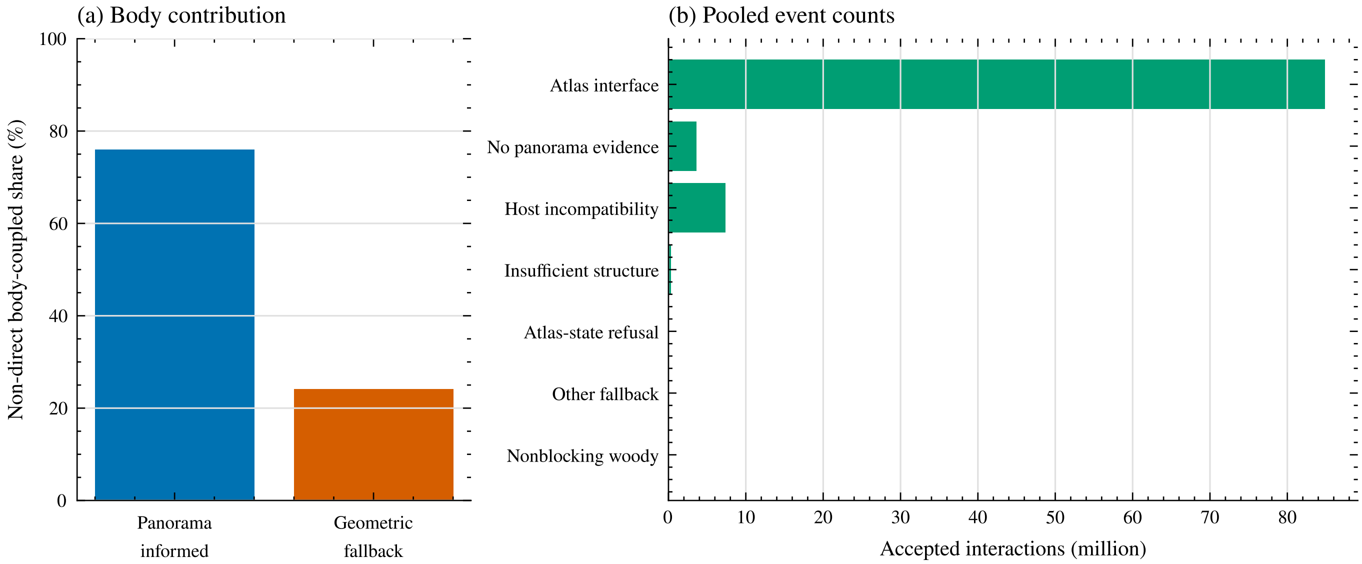

Fig. 1. Ray-reached semantic-evidence coverage. Panel (a) partitions the pooled non-direct body-coupled whole-body SAR contribution between panorama-informed interfaces and declared geometric fallback states. Panel (b) gives accepted retained interaction counts for all seven audit categories. Direct transport is not applicable because it has no material interaction.

TABLE II
SHARE OF POOLED BODY-COUPLED WHOLE-BODY SAR WITHIN EACH RETAINED NON-DIRECT COMPONENT. OTHER FALLBACK COMBINES INSUFFICIENT STRUCTURAL MASS, ATLAS-STATE REFUSAL, AND THE RESIDUAL FALLBACK CATEGORY.

Component	Panorama informed	No panorama evidence	Host incompatible	Other fallback
Order-1 specular	80.643%	8.774%	10.530%	0.052%
First diffuse	13.337%	44.345%	42.314%	0.004%
Combined non-direct	75.903%	11.280%	12.769%	0.049%

The current admission policy also requires complete sky diagnostics and a vertical registration optimum inside the altitude search interval. It refuses a pose with a missing residual, a residual above 4°, missing sky diagnostics, the near inside-geometry signature, or an optimum at the altitude bound. Each atlas stores the exact policy version that selected its cameras. New builds use `registration-admission-v2`. The sealed Korenmarkt atlas uses its recorded version-1 policy and is not silently restamped. An admitted camera can still be skipped if its dense and SAM semantic rasters are missing, incomplete, malformed, or inconsistent with the fixed vocabulary. These refusals prevent an image, pose, model, or mesh mismatch from entering fusion.

III. NUMERICAL CONFIGURATION AND PROVENANCE

The receiver is $1.5\,\mathrm{m}$ above the support surface. Route points are placed at approximately $6\,\mathrm{m}$ arc-length intervals, with registered panorama positions retained as route anchors. The route-visible skyline extraction finds boundary and crease edges that separate the first visible mesh hit from sky. Duplicate mesh-edge contacts are combined. The resulting three-dimensional edge lengths are the quadrature weights. Each source site is shifted vertically by $0.5\,\mathrm{m}$ to avoid self-occlusion. Table IV gives the retained counts.

The first-diffuse term uses a reciprocal next-event estimator. For each primary direction $\mathbf{u}_n$ drawn uniformly over 4π, the tracer keeps the first blocking material vertex. It then draws one roofline site from the arc-length weights and tests the connecting segment. If the connection is visible, its unscaled contribution is

$$w_n = \frac{R_n(1-\kappa_n)}{\pi}\frac{[\mathbf{n}_n\cdot\mathbf{d}_n]_+}{r_n^2}. \tag{1}$$

Here R_n is the unpolarized Fresnel power reflectance, κ_n is the coherent specular share, $\mathbf{d}_n$ points from the vertex to the sampled source, and r_n is the connecting distance. Occluded connections contribute zero. The estimate is $(4\pi/N)\sum_n w_n$. The launch direction determines the physical arrival direction by reciprocity. Only these sampled first-diffuse contributions are accumulated in the 4,096-cell Fibonacci grid. Direct sources and

TABLE III
CONFIGURATION OF THE VERIFIED FIVE-SITE RESULT.

Item	Production value
Sites and routes	Korenmarkt, Prague, Madrid, Mexico City, and Tokyo Hachiko. The fixed routes contain 10, 22, 14, 11, and 16 standpoints.
Geometry and frequency	Original photogrammetric support mesh within a 250 m horizontal radius at 15 GHz. The circular crop area is $196{,}349.54\,\mathrm{m}^2$.
Source measure	Observed route-aligned roofline. Areal density sets the expected source count. Physical three-dimensional roofline length sets the conditional source weights.
Exposure normalization	Per unit $\rho_A P_{\mathrm{EIRP}}$. Normalized whole-body SAR has unit $\mathrm{m}^2\,\mathrm{kg}^{-1}$.
Transport	Exact direct term, exact order-1 all-specular term, and stochastic next-event estimation of the first diffuse interaction at the first blocking material vertex.
Surface model	Atlas material mode with measured finish roughness. The material posterior is evaluated at the exact hit texel.
Monte Carlo	200,000 IID primary rays per standpoint and replica. Seeds 7 through 22 give 16 replicas. Stored convergence looks use 4, 8, 12, and 16 replicas.
Angular output	4,096 passive cells collect only the first-diffuse estimate. Exact direct and order-1 all-specular paths remain directional atoms. The cells are not launch strata.
Body	Duke with 56,024 surface elements, area $1.87250\,\mathrm{m}^2$, mass 72.4 kg, $T_0 = 0.500140$, and fixed route-tangent yaw. Body coupling uses the level-2 surface-field model.

TABLE IV
ROUTE AND ROOFLINE QUADRATURE SIZES.

Site	Route points	Quadrature sites	Length (m)
Korenmarkt	10	457	157.54
Prague	22	502	267.15
Madrid	14	207	79.40
Mexico City	11	164	63.69
Tokyo Hachiko	16	400	248.76

accepted order-1 specular paths are stored as exact directional atoms.

For incidence cosine c, complex relative permittivity ϵ_r, surface root $q = \sqrt{\epsilon_r - (1-c^2)}$, and wavelength λ, the production law is

$$R = \frac{1}{2}\left(\left|\frac{c-q}{c+q}\right|^2 + \left|\frac{\epsilon_r c - q}{\epsilon_r c + q}\right|^2\right), \tag{2}$$

$$\kappa = \exp\left[-\left(\frac{4\pi s c}{\lambda}\right)^2\right]. \tag{3}$$

Thus the reflected specular and diffuse shares are $R\kappa$ and $R(1-\kappa)$. The surface-finish roughness s excludes larger periodic relief. Table V gives the evaluated 15 GHz material table. Metal uses the finite-conductivity value in the same complex-permittivity convention.

The deterministic order-1 search uses a conservative mirrored-receiver triangle-cone broad phase when the candidate set reaches 20,000. Its CUDA Float64 implementation leaves the host exact final test unchanged. The normal adaptive candidate budget is 120 million. One Tokyo standpoint has zero accepted order-1 paths, so a relative stopping rule cannot close around zero. That case records and fully enumerates a 320 million candidate cap. Body coupling uses fixed blocks of 512 directions. Candidate caps, broad-phase thresholds, chunks, and block sizes control computation. They do not change the declared physical model.

The result contains 73 standpoints, 1,168 standpoint-replica fields, 80 site-replica runs, and 233.6 million stochastic primary rays. Every campaign manifest lists 42 files. All 210 recorded file hashes pass. The aggregate JSON, CSV, PDF, and PNG also match their artifact manifest. Across the 1,168 fields, the maximum raw residual between the saved total and the sum of direct, all-specular, and first-diffuse components is $1.735 \times 10^{-18}\,\mathrm{m}^{-2}$. The CUDA body result agrees with the NumPy reference to a maximum relative error of 6.64×10^{-16} in the verified benchmark. The verified artifacts are stored under the roofline campaign output package named `current_five_city_first_material_interaction`. The Duke STL has SHA-256 `781e65ef3882f1347669e0ddca5dafa82cd6368dddd6b9e801dc49613822fe3b`. The sealed body-area array has SHA-256 `5dd410b9512fdb0274dd1ac1fb8f7b7087564abfed998d57a25065e87ffebd6a`. Each calculation manifest also seals the support mesh, atlas, source sites, source weights, route arrays, material tables, and executable configuration.

IV. REPLICA CONVERGENCE AND LOWER TAILS

The 64-replica extension uses seeds 7 through 70 and nested looks of 16, 24, 32, 48, and 64. Its first 16 replicas are exactly equal to the sealed campaign arrays after timing fields are removed. Mexico City standpoints 0, 1, and 3 and Tokyo Hachiko standpoints 13, 14, and 15 remain the explicit shadowed stratum. The first-diffuse estimate is their only nonzero modeled contribution. From 48 to 64 replicas, their largest pointwise whole-body SAR changes are 0.0125 and 0.0104 dB. The route q_{10} changes are 0.00344 and 0.00491 dB, and their whole-replica bootstrap widths are 0.364 and 0.0538 dB.

The bootstrap resamples complete replicas jointly over all standpoints, so it preserves spatial dependence within one replica. Its 2,000 draws use PCG64 with the authenticated analysis seed 20260814. All five identities, manifests, component closures, and common inputs pass, and both lower tails meet the declared 48-to-64 aggregate criteria. Mexico City nevertheless retains rare-event first-diffuse behavior: its maximum positive replica contribution is 5738 times its positive-replica median, compared with 2.38 for Tokyo Hachiko. These intervals remain

TABLE V
EVALUATED TRANSPORT MATERIALS AT 15 GHZ. THE ROUGHNESS COLUMN IS RMS HEIGHT.

Class	$\Re\epsilon_r$	$-\Im\epsilon_r$	s (μm)	Class	$\Re\epsilon_r$	$-\Im\epsilon_r$	s (μm)
Asphalt concrete	4.83	0.569	450	Metal	1.00	1.20×10^7	5
Brick	3.91	0.044	30	Plasterboard	2.73	0.130	150
Ceramic	7.07	0.081	280	Plywood	2.71	0.396	50
Chipboard	2.58	0.215	50	Polymer	1.99	0.103	50
Concrete	5.24	0.461	600	Soil	5.24	0.461	600
Fabric	1.99	0.103	50	Water	5.24	0.461	600
Glass	6.31	0.162	1	Wood	1.99	0.103	50
Marble	7.07	0.081	280				

TABLE VI
CURRENT-CONTRACT CONVERGENCE FROM 48 TO 64 REPLICAS. BOOTSTRAP WIDTH IS THE 95% INTERVAL WIDTH FOR ROUTE q_{10} WHOLE-BODY SAR. SHADOW CHANGE IS THE LARGEST STEPWISE CHANGE AMONG THE SIX DECLARED SHADOWED STANDPOINTS.

Site	q_{10} change (dB)	Bootstrap width (dB)	Shadow change (dB)
Korenmarkt	0.0000723	0.000222	–
Prague	0.0000203	0.000218	–
Madrid	0.00000615	0.000378	–
Mexico City	0.00344	0.364	0.0125
Tokyo Hachiko	0.00491	0.0538	0.0104

conditional on each fixed registered route. They do not include route-selection or city-sampling uncertainty.

V. PRIMARY-RAY AND ANGULAR-CELL BUDGETS

The budget diagnostic used paired seeds and common random numbers at the Madrid, Mexico City, and Prague routes. Exact direct and all-specular components were byte identical in every paired comparison, and the 200,000-ray, 4,096-cell replay matched the sealed baseline. The route-median normalized whole-body-SAR changes were small, with a maximum absolute q_{50} difference of 0.000797345 dB among the five cheaper settings. Directional first-diffuse fields did not meet the stated gate. The largest site q_{90} normalized-L_1 differences were 0.877498, 0.678942, and 0.400218 at 25,000, 50,000, and 100,000 rays, respectively. At 1,024 and 2,048 cells, they were 0.491265 and 0.455431. Each value exceeds the 0.1 gate. Mexico City shadowed-point maximum absolute whole-body-SAR changes were 0.707362, 0.563841, and 0.217259 dB for the three ray settings, compared with 0.00206975 and 0.00088674 dB for the two cell settings.

Ray cuts also do not give a defensible end-to-end speedup. At 25,000 rays, estimator-wall-time ratios relative to the baseline range from 0.991 to 1.064 across the three routes. Exact all-specular work takes 8.06 to 23.79 s in the baseline, whereas stochastic tracing takes 1.15 to 2.40 s. The q_{90} variance-time ratios for every ray-reduced arm exceed one, with a minimum of 3.16. The reported timing therefore leaves the exact specular stage dominant while the reduced ray settings increase first-diffuse variance. Figure 3 retains the 200,000-ray, 4,096-cell setting as the production baseline.

TABLE VII
ATLAS-TO-GEOMETRIC CHANGE IN ROUTE QUANTILES OF NORMALIZED WHOLE-BODY SAR. POSITIVE VALUES MEAN THAT THE ATLAS RESULT IS LARGER.

Site	q10 (dB)	q50 (dB)	q90 (dB)
Madrid	0.233	0.249	0.269
Mexico City	24.84	-0.158	0.104

VI. PAIRED MATERIAL-EVIDENCE CONTROL

The control replaces the panorama-derived atlas layer with the geometric fallback in Madrid and Mexico City. Each pair keeps the support mesh, route, source curve, source weights, body, frequency, transport topology, ray count, output cells, and 16 seeds fixed. A strict compatibility check permits changes only in the material mode, the material arrays and hashes, the run configuration, and the presence of the atlas files. Both paired manifests pass all 42 file hashes. The comparison therefore measures sensitivity to the atlas evidence layer under the current model. It does not isolate reflectance because the atlas also changes roughness, specular share, and the nonblocking state of woody vegetation. It does not measure semantic or material accuracy.

The direct component is identical within each pair. In Madrid, the atlas raises the route-median all-specular whole-body SAR by 1.89 dB and lowers the first-diffuse component by 12.43 dB. Their combined effect changes the total route median by 0.249 dB. Mexico City's 24.84 dB q10 change is set by three shadowed standpoints where both alternatives are near zero and first-diffuse transport is the only nonzero modeled contribution. The Mexico City median and q90 changes are -0.158 and 0.104 dB. The two sites show that component changes can be much larger than the change in total normalized whole-body SAR. The result supports a model-layer sensitivity statement for these routes only.

VII. TIMING BOUNDARY

The repeated numerical campaigns start from a ready city scene. Their wall times on one A6000 are 29.79 s for Korenmarkt, 69.02 s for Prague, 40.70 s for Madrid, 30.92 s for Mexico City, and 50.11 s for Tokyo Hachiko. These times include the 16 transport replicas and body coupling for the complete fixed route. They exclude image and mesh acquisition, panorama registration, semantic inference, depth processing, and atlas construction. Persistent transport caches and exact

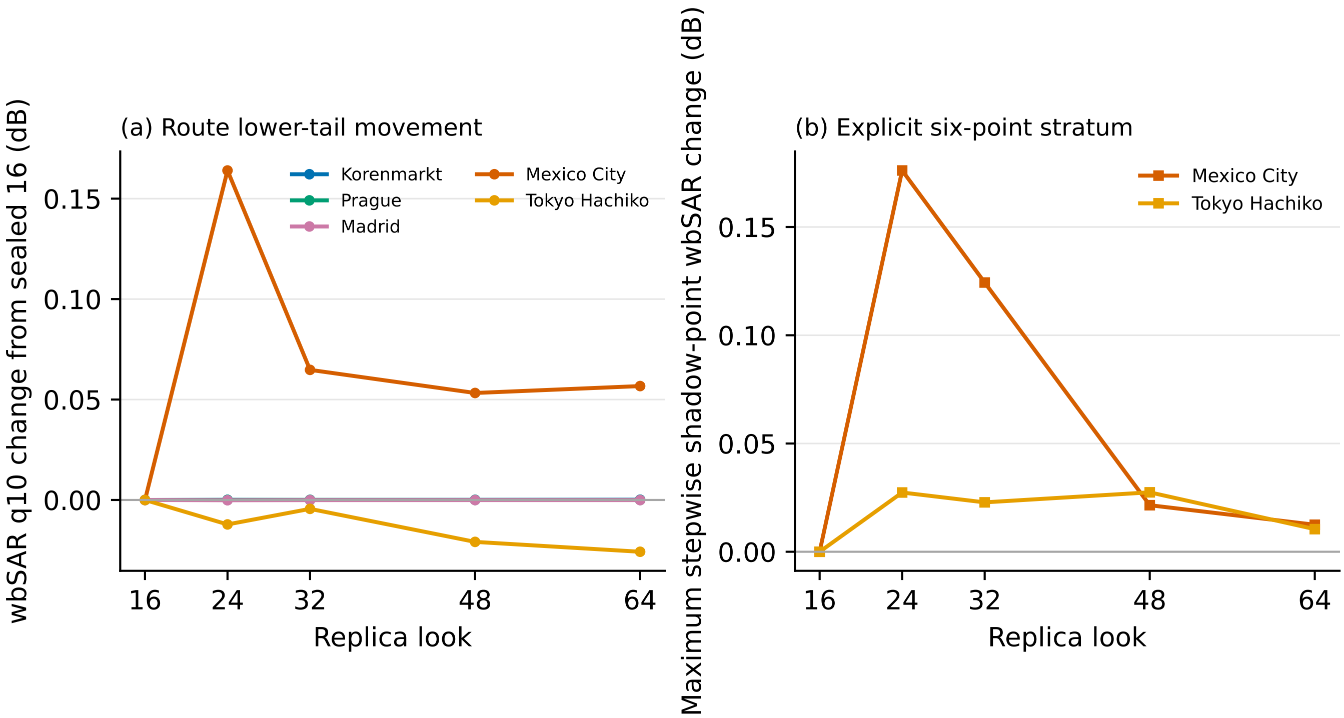

Fig. 2. Current-contract convergence through 64 replicas. Panel (a) shows route q_{10} whole-body SAR changes from the sealed 16-replica value. Panel (b) shows the largest stepwise change among the three shadowed standpoints in each of Mexico City and Tokyo Hachiko. The final 48-to-64 changes satisfy the declared aggregate lower-tail criteria. Mexico City retains rare-event first-diffuse behavior.

conservative broad-phase screening reduce repeated work with no change to the saved scientific arrays.

The cold scene path has no complete common timing ledger. Available 14-panorama records give 21.3 to 26.6 min for the hybrid semantic pass and 5.3 to 13.6 min for atlas construction. Acquisition, registration, depth, and fusion were not all timed separately. One live A6000 parity check took 116.981 s for one panorama with independent model sessions. A shared session processed two panoramas in 199.228 s and reproduced every scientific panorama array, concept cache array, prompt gate, and normalized metadata field exactly. These values are implementation anchors. They do not define a universal time per panorama. A single cold end-to-end time is therefore not reported.

VIII. Additional Limitations

The transport result contains exact direct transport, exact single-reflection all-specular transport, and the first diffuse interaction. A sampled first-diffuse path stops at that diffuse event. The model has no diffuse-to-specular suffix, higher specular order, or complete multipath expansion. The controlled open-square test validates the first-diffuse normalization, visibility, inverse-square loss, and cosine factors. The maximum adjoint-versus-Sionna difference is 0.0621 dB for the bounced term and 0.0344 dB for total transport in that test. The same three-way test has not been repeated for the full five-site stack with its atlas materials and exact specular term. The test provides component validation. It does not provide external validation of every city result. The level-2 body coupling uses one-sided local incidence. It does not trace body self-occlusion.

The five routes are selected case studies. Their differences do not rank the five cities and do not estimate population exposure. Each result is normalized per unit $\rho_A P_{\mathrm{EIRP}}$, so it is not an absolute prediction for an operator deployment. The study uses one 15 GHz frequency, one body phantom, one fixed route-tangent body orientation, one roofline source law, and five plaza routes. Additional frequencies, body models, headings, deployment laws, street canyons, parks, and repeated route selections remain outside the present evidence.

Panorama evidence covers only surfaces that an admitted camera can see and cast onto the support mesh. Unsupported atlas cells use the geometric material. Transient pixels are withheld from the static surface. Image evidence can preserve within-triangle material variation, but it cannot recover geometry that is absent from the photogrammetric mesh. The two-site material control tests downstream sensitivity and supplies no semantic ground truth. Coverage on all support-mesh area also differs from coverage on the subset reached by propagation paths. The ray-reached audit now reports where the recorded atlas and fallback rule act in retained order-1 specular and first-diffuse transport. It does not test whether the panorama labels or fallback material are correct. The current claims therefore remain conditional on those recorded surfaces.

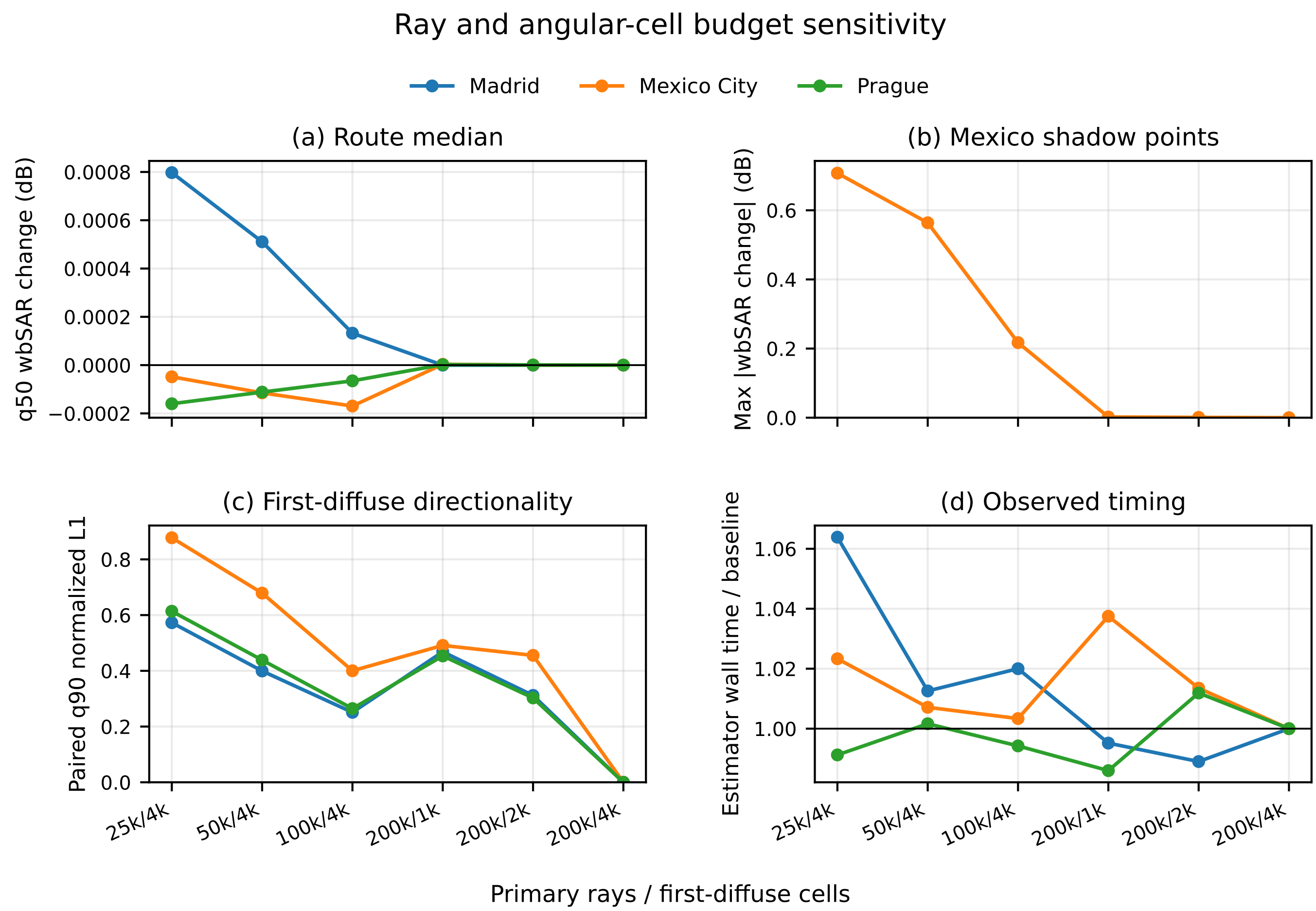

Fig. 3. Primary-ray and angular-cell budget sensitivity under the current first-material transport contract. All five cheaper settings leave route-median whole-body SAR nearly unchanged, but each exceeds the directional first-diffuse gate. Exact direct and order-1 specular components remain invariant. The diagnostic retains 200,000 rays and 4,096 cells.